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CTL Chromebook NL73

Solder-Free Repair Guide



Covers NL73 and NL73-8 clamshells. NL73CT differences are flagged.
Not for NL73T / NL73TW / NL73TWB convertibles.

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Procedures follow CTL's NL72 Self-Repair Manual, which CTL designates as the reference for the NL73, plus CTL's NL73 specs and parts catalog. Confirm NL73-specific details on the first teardown.

1 Read first

This guide covers every NL73 repair the Tech Bench can do with hand tools: swapping a part, reseating a cable, or fixing software. Anything that needs a soldering iron, hot air, or board-level work is listed in Section 6 and goes to CTL or the district lead tech instead.

How this guide was built. CTL's support site lists the NL72 Self-Repair Manual as the reference for the NL73. The screw counts and step order here come from that manual. Parts come from CTL's NL73 catalog, and specs from CTL's NL73 spec sheet. Because the manual was written for the NL72, some repairs have a notes box. Confirm the counts the first time you do each repair on an NL73.

Which NL73 is on your bench? Read the model on the D-cover label. This guide is for the **NL73** (4 GB) and **NL73-8** (8 GB) clamshells. The **NL73CT** (touch clamshell) shares most parts except the screen. **NL73T, NL73TW, and NL73TWB** are convertibles with different keyboards, covers, panels, and hinges. CTL warns that some NL72T parts don't fit the NL73T. Don't use this guide for them.

Bench levels

Level	Who may do it	Typical repairs
LEVEL 1	Any trained student	Software resets, single keys, camera caps, base cover, battery
LEVEL 2	Student with supervisor sign-off on this repair	Display, bezel, cables, USB board, Wi-Fi card, touchpad, camera, hinges, speakers
LEVEL 3	Lead tech or adult supervisor only	Top assembly (keyboard) swap, A cover, liquid damage, mainboard swap

Before you touch any device

1. ☐ Check the asset tag and look the device up in the district inventory. Note any open ticket.
2. ☐ Ask the lead tech whether the unit is still under CTL warranty or an accidental-damage plan. CTL's manual says damage from careless or unsuccessful self-repair isn't covered by the standard warranty, so warranty units may be better sent to CTL.
3. ☐ Photograph all four sides of the device before you open it.
4. ☐ Record the symptom in the user's own words on the ticket.
5. ☐ Warn the user that a reset erases local files (Downloads). Cloud files in Drive are safe.

Safety rules

- **Power off, unplug, remove the microSD card and all cables** (CTL's pre-disassembly steps), then disconnect the battery before any work under the base cover.
- **Lithium batteries:** never puncture, bend, or pry against the battery. A swollen, hot, hissing, or smoking battery is a Level 3 stop. Move it to the metal battery bin or sand and get an adult.
- **ESD:** CTL's manual calls for a wrist grounding strap and conductive mat. Touch boards only by the edges.
- **Ribbon cables:** CTL warns the touchpad and USB-board FFCs can be damaged if pulled while their connectors are locked. Always unlock first.
- **Glass:** wear safety glasses when removing a cracked LCD.
- **Only use CTL-certified parts**, as CTL's manual requires.

Tools and supplies

Item	Use	Notes
Phillips #0 and #1 screwdrivers	All case and board screws	CTL's recommended set
Small flathead screwdriver	Releasing some latches	CTL's recommended set. Never on board components.
Flat plastic pry tool / spudger	Base cover, bezel, connectors	CTL's recommended set
Plastic tweezers	FFC ribbon cables, antenna leads	CTL's recommended set
Wrist grounding strap + conductive mat	Static protection	CTL's recommended set
Magnetic screw mat or labeled tray	Keeping screws in order	Base cover alone has 9 screws
90%+ isopropyl alcohol + soft brush	Cleaning contacts, spill residue	
Kapton tape	Re-taping cables and the keyboard-plate mylar	
Recovery USB or SD card, 8 GB+	ChromeOS recovery	Made with Chromebook Recovery Utility
Known-good USB-C charger (45 W)	Testing power and charging	CTL 45 W USB-C adapter
Safety glasses	Cracked-screen removal	
Fire-safe battery bin or sand bucket	Damaged batteries	Required at every bench

2 Know the machine

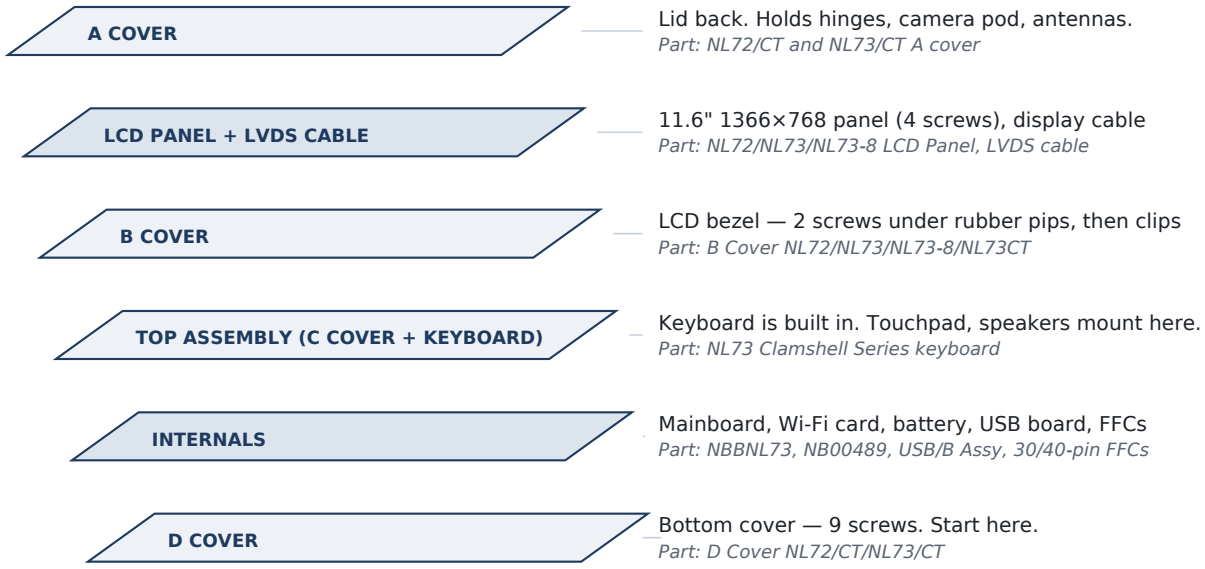
Specs that matter for repair

Item	NL73 / NL73-8	Why it matters
Display	11.6", 1366×768, non-touch (NL73CT: touch)	NL72/NL73/NL73-8 panel fits. NL73CT needs the touch panel.
Processor	Intel N100, quad-core	Soldered to mainboard
Memory	NL73: 4 GB • NL73-8: 8 GB LPDDR5	Soldered. Determines which mainboard to order.
Storage	64 GB eMMC	Soldered. Storage failure = mainboard.
Battery	Internal, one battery fits the whole NL73 series	NBBNL73. CTL doesn't publish its Wh rating.
Ports	2× USB-C, 2× USB-A, microSD, audio	One USB-C and the microSD slot sit on the USB board, per CTL's manual figures
Wireless	Wi-Fi 6E (Intel AX211), Bluetooth 5.3	Socketed card, 1 screw. Not the NL72's AX201.
Camera	720p HD, rotating camera pod	Module plus front cap, back cap, and hinge cap
Keyboard	Spill- and drop-resistant, anti-peel keys	Built into the top assembly. Single keys also sold.
Hinge	180° clamshell	Left and right sold separately
Durability	MIL-STD-810H, 75 cm drop, carry handle	
Weight	2.84 lb	
Updates	ChromeOS auto-updates to June 2033	
SKU	NL73 CBUS1100028 • NL73-8 CBUS1100024 • NL73CT CBUS1100026	Read the D-cover label

Specs from CTL's NL73 series spec sheet. The USB board's port assignments are inferred from the NL72 manual's figures. Confirm on teardown.

Anatomy: the A/B/C/D covers

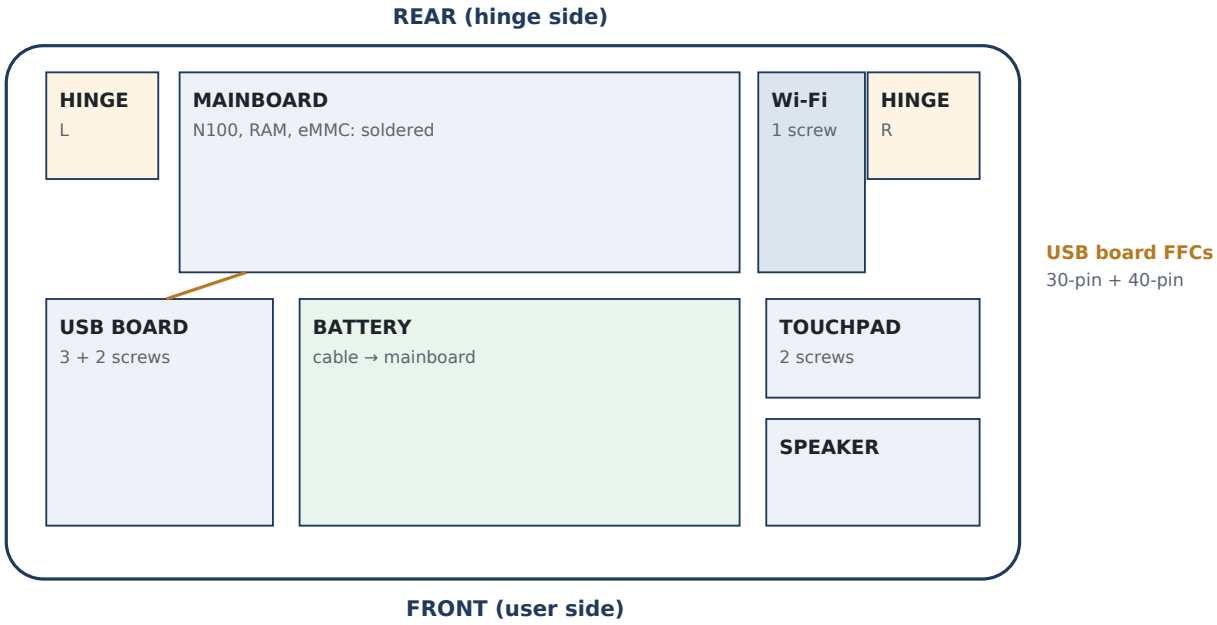
CTL names the shell pieces by letter, and parts are ordered by these names. On the NL73 the keyboard is part of the C cover, called the **top assembly**.



Top of stack = lid exterior. Bottom of stack = underside. Screw counts from CTL's NL72 manual, which CTL says to use for the NL73.

Inside the base

With the D cover off, you see the parts below. Wi-Fi sits at the right side of the mainboard, per CTL's manual.



ILLUSTRATIVE — screw counts are from CTL's NL72 manual; block positions are approximate.

NL73 parts list

Parts CTL lists as fitting the NL73 clamshell, as of October 2026. Stock changes often. Items marked † are NL72 listings that don't name the NL73. Ask CTL to confirm fit before ordering. Order at ctl.net/collections/chromebook-parts.

Part (CTL listing name)	SKU	Repair
Replacement Battery for NL73 Series (all NL73 models)	NBBNL73	R4
NL73 Clamshell Series keyboard (NL73, NL73-8, NL73CT)	NBKEYNL73	R17
US individual key replacement, NL73 series	—	R1
NL72, NL73 and NL73-8 LCD Panel (non-touch)	—	R11
NL72 and NL73 LVDS Cable (clamshell)	—	R12
B Cover NL72/NL73/NL73-8/NL73CT	—	R10
A cover NL72/CT and NL73/CT	—	R15
D Cover NL72/CT/NL73/CT	—	R3
D bottom cover screws, NL72 series†	—	R3
Left Hinge / Right Hinge NL72/CT/L/NL73/CT	—	R14
Camera Module NL72/NL72L/NL73/NL73-8/NL73CT	NB00430	R13
Front Camera Cap, Back Camera Cap, Camera Hinge (cap)	—	R2
USB/B Assy for NL73 Clamshell Series (listed as "USD/B")	—	R8
30-pin and 40-pin USB/B ribbon cables	—	R9
Touch Pad, NL72 series†	—	R7
Left / Right Speaker, NL72†	—	R16
Mainboard NL73 4/64	NB00489	R19
Mainboard NL73-8 and NL73CT (8/64)	—	R19
CTL 45 W USB-C Adapter	—	Triage

Don't use the NL72's Wi-Fi card. CTL sells an AX201 Wi-Fi module for the NL72. The NL73 ships with an AX211 (Wi-Fi 6E). Ask CTL for the correct NL73 card rather than fitting the NL72 part.

Stocking suggestion. Keep camera caps, individual keys, D-cover screws, and both USB ribbons on hand. They're small, cheap, and cover the most common tickets.

3 Triage: symptom → repair

Start with the cheapest, least invasive fix. Work through the "try first" column before ordering parts.

Symptom	Try first	Then	Part likely needed
Won't turn on, no lights	Known-good charger 30 min → S1	R5 reseat battery → R4	Battery, or mainboard
Not charging	Other USB-C port, other charger → S1	S2 battery test → R4 → R8 if one port only	Battery, USB board, or mainboard
Battery drains fast / low health	S2 battery test	R4	Battery
Swollen battery / bulging base	STOP — Level 3	R4 by lead tech	Battery
Black screen, power light on	External display via USB-C; S1	R5 → R12 → R11	LVDS cable or panel
Flicker or lines that change with lid angle	S1	R12 LVDS cable	LVDS cable
Cracked or blotchy screen	—	R11 LCD panel	LCD panel
Bezel loose or broken	Reseat clips, check 2 screws	R10	B cover
Lid cracked	—	R15	A cover
Lid floppy, hinge cracked	Check hinge screws	R14	Hinge(s)
Camera cap missing, camera pod loose	—	R2	Camera caps
Camera not detected	S1, check camera pod rotation	R13	Camera module
One key missing or broken	—	R1	Individual key
Several keys or a row dead	S2 keyboard test; S1	R5 reseat keyboard FPC → R17	Top assembly
Touchpad dead or erratic	S1; check touchpad settings	R5 → R7	Touchpad or its FFC
USB-A, microSD, or one USB-C port dead	Test with another device; S1	R9 ribbons → R8 USB board	Ribbons or USB board
No sound	Volume, output setting; S1; S3	R5 speaker cable → R16	Speakers
Wi-Fi missing or weak	S1 → S3	R6 reseat card and antennas	Wi-Fi card (ask CTL)
Liquid spill	Power off NOW, unplug	R18 (Level 3)	Depends on damage
Slow, crashing, errors	S1 → S3	Sec. 6 if recovery fails	Often none
"ChromeOS is missing or damaged"	S3 recovery	R19 if recovery can't write	Mainboard
Base cover cracked or screws missing	—	R3	D cover or screws

4 Software repairs

These need no tools and clear many tickets. Always do S1 before opening a device.

S1 Hard reset (EC reset) LEVEL 1

Fixes	Won't power on, won't charge, frozen, black screen, unresponsive keyboard or touchpad
Time	2 minutes
Tools	Charger
Data loss	None

CTL's manual says to try a power cycle before any reset. A hard reset restarts the embedded controller, which manages power, keyboard, and charging.

Procedure

1. ☐ Unplug the charger and any USB devices. Remove the microSD card.
2. ☐ Hold Power for 10 seconds so the device is fully off.
3. ☐ Hold **Refresh** (circular arrow), then tap **Power**. Release Refresh when it starts.
4. ☐ If nothing happens, plug in a known-good 45 W USB-C charger, wait 30 minutes, then repeat steps 2–3 with the charger connected. Try the other USB-C port too.

Verify before closing the ticket

- Device reaches the login screen.
- Charging light comes on with the charger connected.

Tip. After the battery has been disconnected (R4, R5), plug in a charger before deciding a device is dead. Many Chromebooks won't wake on the power button alone.

S2 Built-in hardware diagnostics LEVEL 1

Fixes	Confirms the fault before you order a part: battery, CPU, memory, keyboard, touchpad, network
Time	5–15 minutes
Tools	None
Data loss	None

Procedure

1. ☐ Sign in with a bench test account if you can.
2. ☐ Open the **Launcher**, search **Diagnostics**, and open it.
3. ☐ **System:** read battery health and cycle count, then run the battery charge and discharge tests.
4. ☐ Run the CPU and memory tests.
5. ☐ **Connectivity:** confirm Wi-Fi is detected and run the tests.
6. ☐ **Input:** open the keyboard tester and press every key. Test the touchpad.
7. ☐ Screenshot any failed result (Ctrl + Show windows) and attach it to the ticket.

Verify before closing the ticket

- Results recorded on the ticket.
- Parts ordered only for a failed test or a fault you saw yourself.

S3 Powerwash and USB recovery

LEVEL 1

Fixes	Slow, crashing, corrupted OS, "ChromeOS is missing or damaged", outdated OS
Time	Powerwash 10 min; recovery 20–40 min
Tools	Recovery USB or SD card (8 GB+)
Data loss	**Yes.** Recovery permanently erases all local storage.

District-managed devices. A powerwash or recovery on an enrolled device should force it to re-enroll on its first Wi-Fi connection. Confirm with the lead tech that forced re-enrollment is on and the bench Wi-Fi can reach enrollment. Never enter Developer Mode on a district device.

Procedure

1. ☐ **Check for updates first:** Settings > About ChromeOS > Check for updates. An outdated OS causes many odd faults.
2. ☐ **Powerwash:** sign out, press Ctrl + Alt + Shift + R, select Restart, then Powerwash > Continue. (May be blocked by policy on managed devices. If so, use recovery.)
3. ☐ **Make recovery media:** on a working computer, install the Chromebook Recovery Utility, open it, and find the CTL NL73 by searching its model name or picking from the list. Insert an 8 GB+ USB drive or SD card and click Create now.
4. ☐ On the NL73, hold **Esc + Refresh**, tap **Power**, and release the keys when the recovery message appears.
5. ☐ Insert the recovery media and follow the prompts.
6. ☐ After it reboots, connect to Wi-Fi and confirm the device re-enrolls to the KPS domain.

Verify before closing the ticket

- Device boots to the enrollment or KPS sign-in screen.
- Shows as enrolled in the Admin console.
- If recovery can't write to storage, the eMMC has failed: R19 or Sec. 6.

Recovery error messages, from CTL's manual: **"An unexpected error has occurred"** — erase the media and retry, or use a different drive. **"Please remove all connected devices"** — unplug everything and restart recovery. **"Does not contain Chrome OS"** — wrong model image or bad drive. **"Security module not working"** — escalate to the lead tech. The NL72 manual's recovery model code doesn't apply to the NL73: choose the NL73 in the utility, or ask CTL for the image.

5 Hardware repairs

Most hardware repairs start with R3 (base cover off) and the first steps of R4 (battery disconnected). They're written out once and referenced everywhere else. Screw counts in **bold** come from CTL's NL72 manual.

> **Golden rules.** Photograph every layer before removing it. Lay screws on the mat in the same pattern as the device. Unlock every connector before pulling a cable. Never force a part: if it won't move, find the screw or clip you missed. Test before the last screws go in.

R1 Individual key replacement — LEVEL 1

Fixes	Missing, cracked, or loose key
Part	US individual key replacement for the NL73 series
Time	5 minutes
Tools	Plastic pick, tweezers

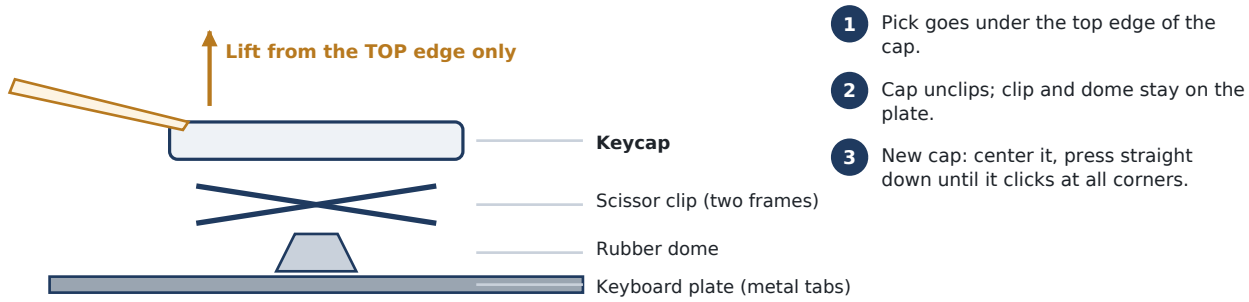


Illustration: side view of one key. Not to scale. The NL73 has anti-peel keys: go gently.

The NL73 has anti-peel keys, built to resist being pried off. Work gently and lift from the top edge only.

Procedure

1. ☐ Power off. Open the lid flat on the mat.
2. ☐ Slide a plastic pick under the top edge of the damaged key and lift until it unclips.
3. ☐ Inspect the scissor clip and rubber dome. If the clip is broken, ask CTL whether the replacement key includes one.
4. ☐ To fit a clip: hook its pins under the metal tabs on the keyboard plate, then press it flat.
5. ☐ Center the new key over the clip and press straight down until it clicks at every corner.
6. ☐ Wide keys (Space, Shift, Enter, Backspace): seat the metal stabilizer wire in its holders first.

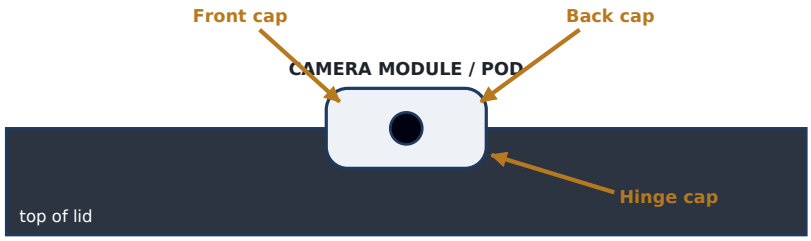
Verify before closing the ticket

- Key clicks evenly and springs back.
- Key registers in the S2 keyboard tester.

R2 Camera caps and camera hinge cap

LEVEL 1

Fixes	Missing, cracked, or loose plastic around the rotating camera
Part	Front Camera Cap, Back Camera Cap, or Camera Hinge (cap)
Time	10–15 minutes
Tools	Plastic picks



1. Bezel off first (it covers the camera's front).
2. Unplug the small camera cable with tweezers.
3. Lift the module by its board edges, never the lens.
4. Test in the Camera app before closing up.

Illustration: camera at the top-center of the lid. Cap shapes simplified.

The NL73's camera sits in a pod at the top of the lid. Three small plastic parts cover it, and students bump and knock them off often. Some may require the bezel off (R10). Confirm on your first one and record it below.

Procedure

1. ☐ Power off and unplug.
2. ☐ Look closely at which cap is damaged: front (screen side), back (lid side), or the hinge cap the pod turns on.
3. ☐ Pry the damaged cap off with a plastic pick, starting at a seam. If it won't release, remove the bezel (R10) and check for screws or tabs from behind.
4. ☐ Check the camera module and its cable for damage. If either is damaged, do R13 now.
5. ☐ Line up the new cap and press it until every clip seats.

Verify before closing the ticket

- Cap is flush with no gap.
- Camera pod rotates fully and holds position.
- Camera app shows a clear image both ways.

First-teardown notes — fill in the first time you do this repair

Front cap: clips only, or bezel off? _____

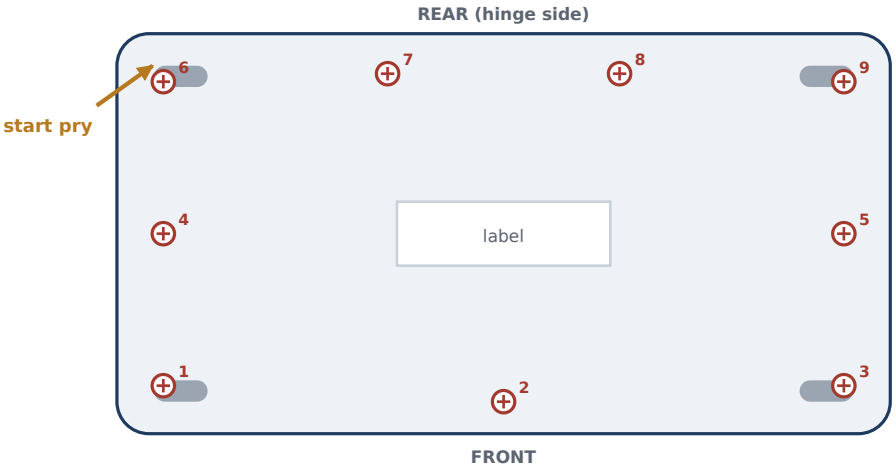
Back cap: clips only, or bezel off? _____

Camera hinge cap: how removed? _____

R3 Base cover (D cover) removal and replacement

LEVEL 1

Fixes	Cracked or missing D cover, missing screws. Also the first step for R4–R19.
Part	D Cover NL72/CT/NL73/CT • D-cover screws (sold individually)
Time	10 minutes
Tools	Phillips #0 or #1, plastic pry tool, screw mat



1. Remove all 9 screws (CTL count).
2. Lay screws on the mat in this same pattern.
3. Pry from the corner shown, then work around the edge.
4. Disconnect the battery before anything else.

Illustration: underside. 9 screws per CTL's manual; exact positions illustrative.

Procedure

1. ☐ **Pre-disassembly (from CTL's manual):** place the device on a stable surface. Remove the charger, the microSD card, and all cables.
2. ☐ Close the lid and flip the device over onto the mat.
3. ☐ Remove the **nine (9)** base-cover screws. Place each on the screw mat in its position.
4. ☐ Starting at a corner, pry the cover open with your thumb and forefinger, a pry tool, or a spudger.
5. ☐ Work around the edge to release the clips, then lift the cover off.
6. ☐ **Stop and disconnect the battery (R4 steps 2–3) before anything else.**
7. ☐ To refit: line up the cover, press around the edge until every clip snaps, then refit all 9 screws. Snug only.

Verify before closing the ticket

- No gaps around the seam.
- All 9 screws refitted. Device sits flat.

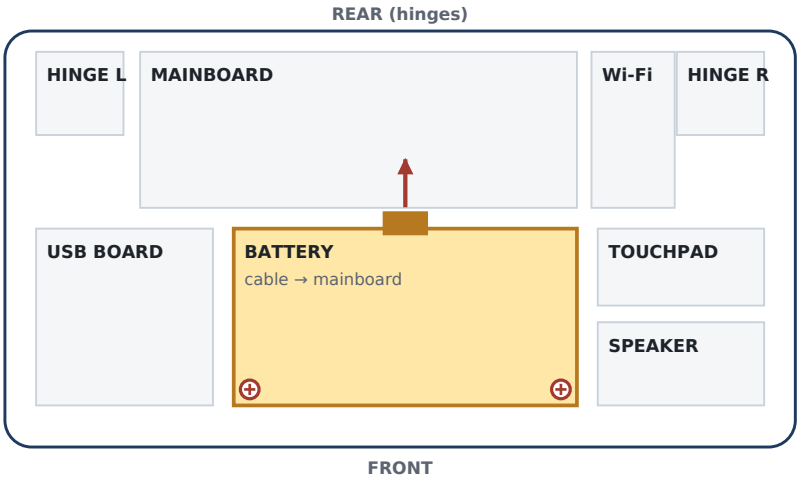
First-teardown notes — fill in the first time you do this repair

Screwdriver size that fits	_____
Screw lengths (all the same?)	_____
Hidden screws under feet? (Y/N)	_____

R4 Battery replacement

LEVEL 2

Fixes	Won't hold charge, low health in S2, swollen battery
Part	Replacement Battery for NL73 Series, NBBNL73
Time	15–20 minutes
Tools	Phillips driver, plastic spudger, ESD strap



1. Unplug the battery cable from the mainboard first (red arrow).
2. Pry at the plug body, never the wires.
3. Remove the battery screws (count them).
4. Lift out flat. Never bend or pry against the cell.

Illustration: base interior, battery highlighted. Screw positions illustrative.

Swollen battery = Level 3. Don't press or pry. The lead tech handles it, and the old cell goes straight to the fire-safe bin.

Procedure

1. ☐ Do R3 to remove the base cover.
2. ☐ Find the battery and its cable. Per CTL's manual, the cable runs from the battery to a connector on the mainboard.
3. ☐ Disconnect the battery cable from the mainboard connector. Pry at the connector body, never the wires.
4. ☐ Remove the screws holding the battery. CTL's manual doesn't list how many. Record it below.
5. ☐ Lift out the battery. Never bend it.
6. ☐ Fit the new battery, refit its screws, and reconnect the cable firmly.
7. ☐ Refit the D cover (R3 step 7).
8. ☐ Plug in the charger before powering on.

Verify before closing the ticket

- Charging light comes on.
- Device powers on and shows a battery percentage.
- S2 shows near-100% health on the new battery.
- Old battery goes to recycling. CTL offers free battery recycling through Call2Recycle.

First-teardown notes — fill in the first time you do this repair	
Number of battery screws	_____
Connector type: lift-up or slide-out?	_____
Adhesive under battery? (Y/N)	_____

R5 Reseating internal connectors

LEVEL 2

Fixes	Dead keyboard, touchpad, or speakers; black screen; dead ports; no power after a drop
Part	None, unless a cable is damaged
Time	15–25 minutes
Tools	Plastic spudger, plastic tweezers, isopropyl alcohol

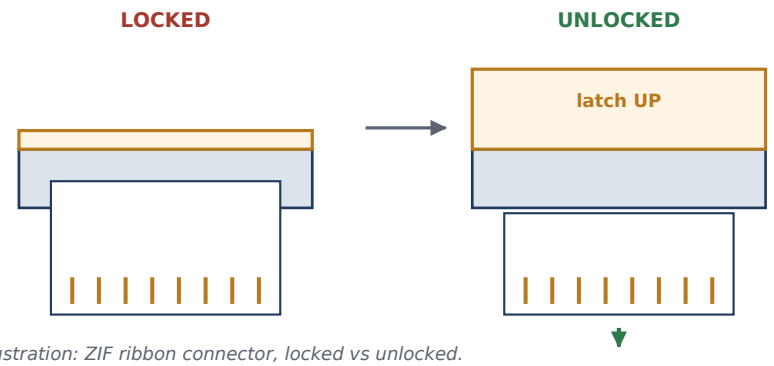


Illustration: ZIF ribbon connector, locked vs unlocked.

1. Flip the hinged latch up (or slide a drawer-type latch out) with a spudger.
2. Ribbon now slides out with zero force. Never pull a locked ribbon.
3. Refit: contacts facing the same way, push in straight and fully, close latch.
4. Board-to-wire plugs (battery, speaker): pry at the plug body, never the wires.

Connector (from CTL's manual)	Type	How to release
Battery cable	Board-to-wire	Pry at the connector body
LVDS (display) cable	Board-to-wire with locking arm	Remove the 2-screw protective plate, lift the locking arm, then pull
Touchpad FFC	ZIF ribbon	Unlock the latch first or the cable is damaged
Keyboard FPC	ZIF ribbon	Unlock the latch first
USB board FFCs (30-pin, 40-pin)	ZIF ribbon	Unlock both ends first
Speaker cable	Board-to-wire	Pry at the connector body
Wi-Fi antennas (2)	Snap-on coax	Lift straight up with tweezers

Procedure

1. ☐ Do R3 and disconnect the battery.
2. ☐ Hold Power for 5 seconds with the battery disconnected to drain leftover power.
3. ☐ For each connector related to the symptom: photograph it, unlock and release it, and inspect it for bent pins, burn marks, or creases.
4. ☐ Clean dirty or corroded contacts with a brush dampened in isopropyl alcohol. Let them dry for 5 minutes.

5. ☐ Reconnect each one. Ribbons must go in straight and all the way before you close the latch. A crooked ribbon is the #1 cause of "still broken."
6. ☐ Reconnect the battery last. Plug in and test **before** refitting the D cover.

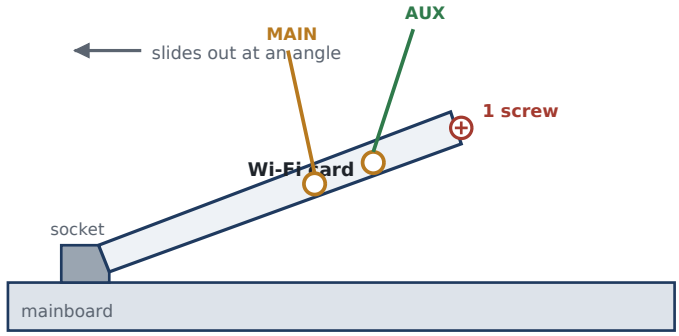
Verify before closing the ticket

- Symptom gone in a full test, not just at boot.
- Creased or torn ribbons replaced (R9, R12).

R6 Wi-Fi card reseat or replacement

LEVEL 2

Fixes	No Wi-Fi or Bluetooth, weak signal, Wi-Fi drops after a drop
Part	Usually none (reseat). Replacement: ask CTL for the NL73's AX211 card. Don't use the NL72's AX201.
Time	15 minutes
Tools	Phillips #0, plastic tweezers



1. Photograph which antenna goes on which terminal.
2. Lift both antenna plugs straight up with plastic tweezers.
3. Remove the 1 screw. The card springs up at an angle.
4. Slide it out of the socket. Refit in reverse.

Illustration: M.2 Wi-Fi card, side view. Colors of antenna leads vary.

CTL's manual lists the WLAN module as self-repairable. It's a socketed card held by one screw, with two antenna leads.

Procedure

1. ☐ Do R3 and disconnect the battery.
2. ☐ Find the Wi-Fi card at the right side of the mainboard.
3. ☐ Note which antenna lead goes on which terminal (photo). Lift both leads straight off with plastic tweezers.
4. ☐ Remove **one (1)** screw.
5. ☐ The card tilts up. Slide it out of the mainboard socket.
6. ☐ To reseat or replace: slide the card into the socket at an angle, press it down, refit the screw, and snap both antennas on in the same positions.
7. ☐ Reconnect the battery and test.

Verify before closing the ticket

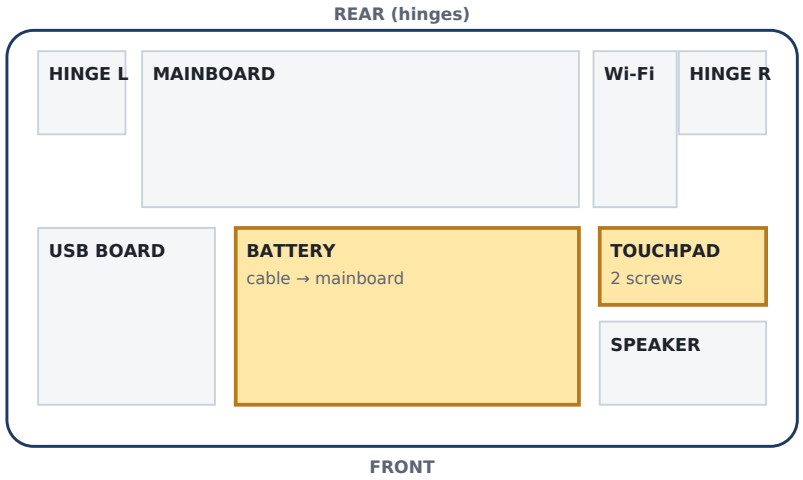
- Wi-Fi connects and S2 connectivity tests pass.
- Bluetooth finds a device.

- Weak signal after a lid repair usually means an antenna lead got pinched in the hinge. Check the routing.

R7 Touchpad replacement

LEVEL 2

Fixes	Touchpad dead, erratic, or physically damaged, with the keyboard working
Part	Replacement Touch Pad for CTL NL72 Series (confirm NL73 fit with CTL)
Time	25 minutes
Tools	Phillips #0, plastic tweezers, Kapton tape



1. Remove the battery first (CTL prerequisite).
2. Unlock and unplug the touchpad FFC at the touchpad end.
3. Peel back the keyboard-plate mylar; remove 2 screws.
4. Peel the grounding strip off the touchpad (leave it on the plate).

Illustration: base interior. Touchpad sits under the battery area; position illustrative.

Check first: touchpad enabled in Settings, S1 done, and R5 reseat tried.

Procedure

1. ☐ Do R3 and remove the battery (R4 steps 2–5). CTL lists battery removal as a prerequisite.
2. ☐ Find the touchpad module and its FFC.
3. ☐ Unlock the latch, then disconnect the touchpad FFC from the touchpad connector. Leave the other end on the mainboard. **Don't pull a locked FFC.**
4. ☐ Peel back the keyboard plate mylar. Keep it intact to restick.
5. ☐ Remove **two (2)** screws. Disconnect the remaining cable at the touchpad connector. Peel back the grounding strip, leaving it attached to the metal plate.
6. ☐ Remove the touchpad module.
7. ☐ Fit the new touchpad, refit 2 screws, restick the grounding strip and mylar, and reconnect the FFC and latch.
8. ☐ Refit the battery, reconnect, and test before closing.

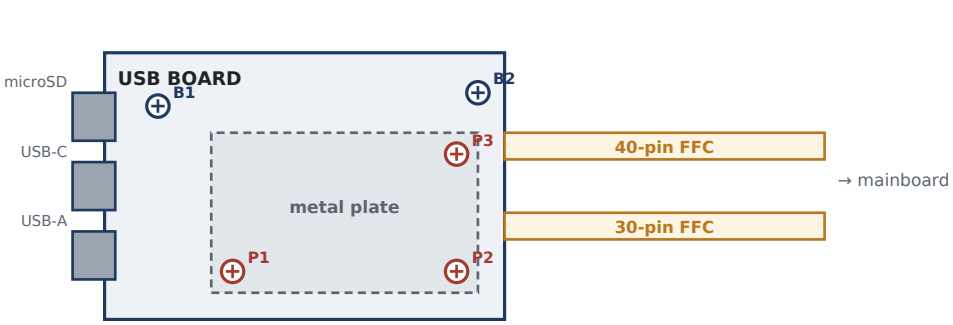
Verify before closing the ticket

- Cursor moves, clicks, and two-finger scrolls.
- Grounding strip reattached. Without it, the touchpad can behave erratically.

R8 USB board replacement

LEVEL 2

Fixes	Dead USB port or microSD slot on the USB board side; one USB-C port won't charge
Part	USB/B Assy for the NL73 Clamshell Series
Time	25–40 minutes
Tools	Phillips #0/#1, plastic tweezers



1. Remove the 3 plate screws (P), then the plate.
2. Remove the 2 board screws (B).
3. Unlock both connectors, then slide out the 30- and 40-pin ribbons.
4. Lift the board, tilting ports out of the case.

Illustration: USB board from above. Port order and screw spots illustrative. Confirm on yours.

Try R9 first: a ribbon is cheaper than the board. CTL's manual lists the LCD module as a prerequisite, so the lid may need to come off to reach this board. Check on your first teardown whether you can reach the screws with the lid attached.

Procedure

1. ☐ Do R3 and disconnect the battery.
2. ☐ If the board's screws or plate are under the hinge area, remove the top half of the unit (R14 steps 2–5).
3. ☐ Find the USB board.
4. ☐ Remove **three (3)** screws from the protection plate, then remove the plate.
5. ☐ Remove the remaining **two (2)** screws on the board.
6. ☐ Unlock both connectors, then disconnect the 30-pin and 40-pin FFCs.
7. ☐ Remove the board. Fit the new one in reverse: 2 board screws, both FFCs locked, plate with 3 screws.
8. ☐ Reconnect the battery and test every port before closing.

Verify before closing the ticket

- Every port on the board works: USB-A reads a drive, USB-C charges, microSD reads.
- Plate refitted with all 3 screws.

First-teardown notes — fill in the first time you do this repair

Which ports are on the USB board? _____

Did the lid have to come off? (Y/N) _____

R9 USB board ribbon cables (30-pin and 40-pin) LEVEL 2

Fixes	Ports on the USB board dead or intermittent, especially after a drop
Part	30-pin USB/B ribbon • 40-pin USB/B ribbon
Time	15–20 minutes
Tools	Plastic tweezers, Kapton tape

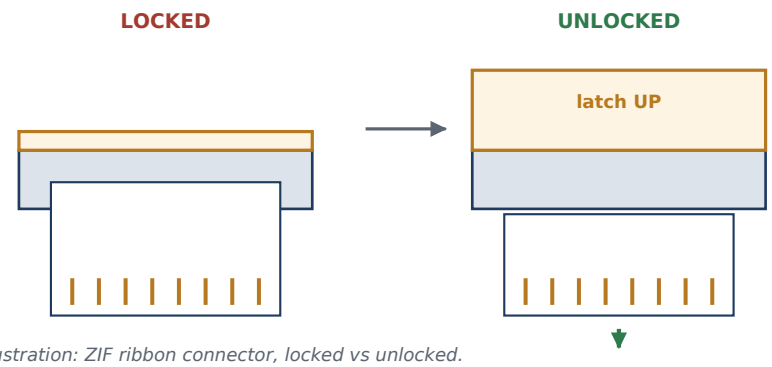


Illustration: ZIF ribbon connector, locked vs unlocked.

1. Flip the hinged latch up (or slide a drawer-type latch out) with a spudger.
2. Ribbon now slides out with zero force. Never pull a locked ribbon.
3. Refit: contacts facing the same way, push in straight and fully, close latch.
4. Board-to-wire plugs (battery, speaker): pry at the plug body, never the wires.

Procedure

1. ☐ Do R3 and disconnect the battery.
2. ☐ Photograph each ribbon's route and tape.
3. ☐ Unlock the connectors at both ends (USB board and mainboard), then slide the ribbon out.
4. ☐ Match the replacement: 30-pin or 40-pin.
5. ☐ Slide the new ribbon in at both ends, contacts the same way as the old one, and lock both latches.
6. ☐ Re-tape and test before closing.

Verify before closing the ticket

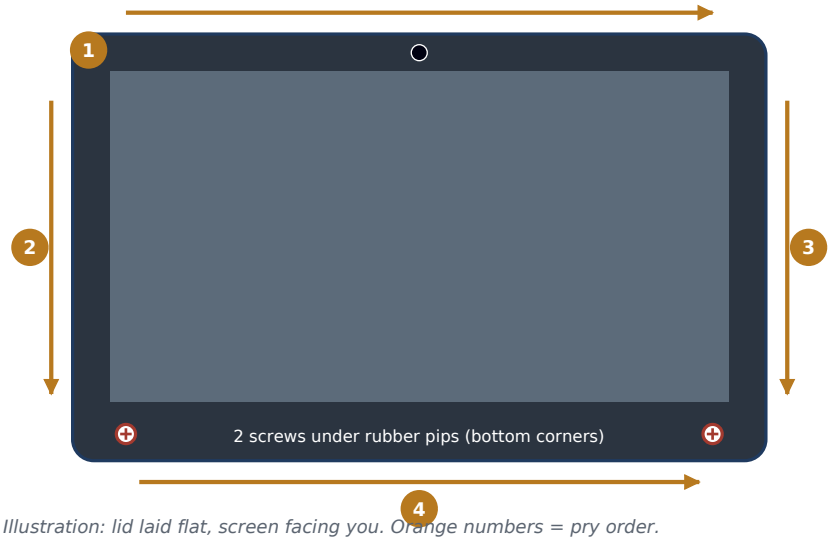
- All USB-board ports work.
- Ribbons lie flat without a kink.

Replace both ribbons at once. They're cheap, and it saves reopening the device.

R10 LCD bezel (B cover) replacement

LEVEL 2

Fixes	Cracked, loose, or missing bezel. First step for R11, R12, R13.
Part	B Cover for NL72/NL73/NL73-8/NL73CT
Time	10 minutes
Tools	Phillips #0, flat pry tool, small pick



1. Remove the pips and the 2 bezel screws.
2. Start at the top-left corner with a flat pry tool.
3. Work down the left side, across the top, down the right side.
4. Finish along the bottom (hinge) edge last.
5. Don't pull on the camera window area.

Illustration: lid laid flat, screen facing you. Orange numbers = pry order.

Procedure

1. ☐ Power off and unplug. Open the lid flat.
2. ☐ Pick out the rubber **pips** covering the **two (2)** bezel screws, then remove both screws.
3. ☐ Pry from the **top-left corner** with a flat pry tool or spudger.
4. ☐ Work down the left side, then the right side, toward the hinge.
5. ☐ Continue along the bottom edge to release the bezel. Lift it away.
6. ☐ To fit: press the bezel on around the edges until every clip snaps, refit 2 screws, and press the pips back in.

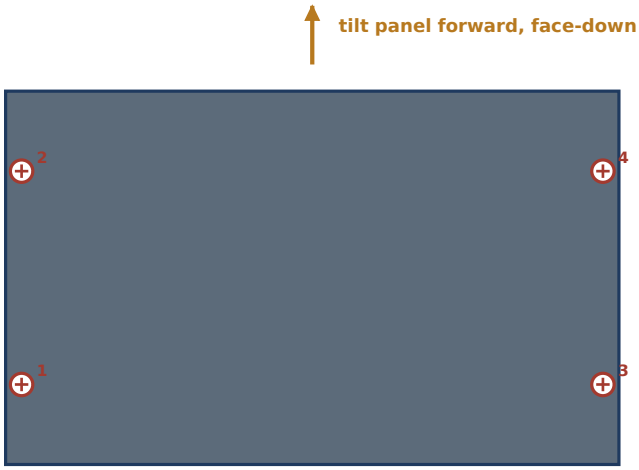
Verify before closing the ticket

- Bezel flush all around.
- Both pips in place (students pick at loose ones).
- Lid closes fully.

R11 LCD panel replacement

LEVEL 2

Fixes	Cracked screen, dead pixels, blots, black screen not fixed by the cable
Part	NL72, NL73 and NL73-8 LCD Panel. **NL73CT needs the touch panel.**
Time	25 minutes
Tools	Phillips #0, plastic picks, safety glasses, Kapton tape



1. Remove the 4 panel screws (sides of the panel).
2. Lay the panel face-down on the keyboard (cloth under it).
3. On the back: peel the tape over the connector, open the clip, pull the cable straight out.
4. Connect the new panel and TEST before screwing it down.

Illustration: lid with bezel removed. Screw positions are typical. Count them on yours.

Cracked glass. Wear safety glasses. Tape across a shattered panel before handling it.

Test first: if an external USB-C display works, the mainboard is fine and the fault is the panel or the cable.

Procedure

1. ☐ Do R3 and disconnect the battery.
2. ☐ Do R10 to remove the bezel.
3. ☐ Remove the **four (4)** LCD panel screws.
4. ☐ Tilt the panel forward, face down onto a cloth on the keyboard.
5. ☐ Remove the sticky tape over the connector, open the clip, and remove the LVDS cable from the panel.
6. ☐ Connect the cable to the new panel, close the clip, and re-tape.
7. ☐ Reconnect the battery and power on to check the picture **before** screwing the panel down.
8. ☐ Power off, disconnect the battery, and refit 4 screws, the bezel, and the D cover.

Verify before closing the ticket

- Full image with no lines, dead pixels, or uneven backlight.
- Brightness works across the full range.
- Image steady through full lid travel.

R12 LVDS (display) cable replacement

LEVEL 2

Fixes	Flicker, lines, or black screen that changes with lid angle. External display works.
Part	NL72 and NL73 LVDS Cable (Clamshell). **NL73CT uses a different cable.**
Time	40–50 minutes
Tools	Phillips #0, plastic tweezers, Kapton tape

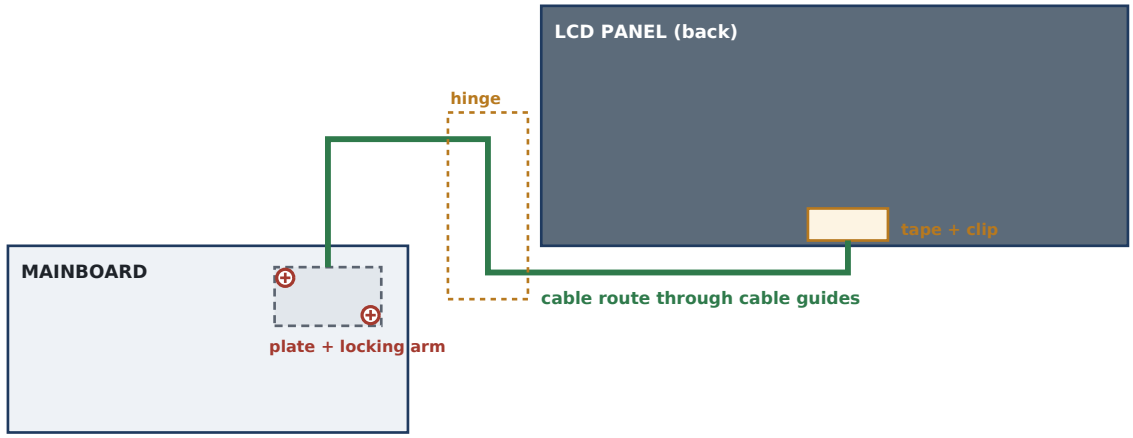


Illustration: display cable path, mainboard to panel (route simplified). Photograph the real route before removing.

Procedure

1. ☐ Do R3 and disconnect the battery.
2. ☐ At the mainboard: remove the **two (2)** screws on the protective plate, lift the locking arm, and remove the LVDS cable.
3. ☐ Photograph the route, then unthread the cable from the cable guides.
4. ☐ Do R10 (bezel) and R11 steps 3–5 (panel face down, cable unplugged).
5. ☐ Free the cable from the lid and hinge area. Note whether the camera is wired through it.
6. ☐ Route the new cable exactly along the old path, threaded through every guide.
7. ☐ Connect both ends: panel clip closed and taped, mainboard locking arm down, plate refitted with 2 screws.
8. ☐ Reconnect the battery and test before reassembling.

Verify before closing the ticket

- Steady image at every angle. Open and close the lid 20 times while watching.
- Camera still works.

First-teardown notes — fill in the first time you do this repair

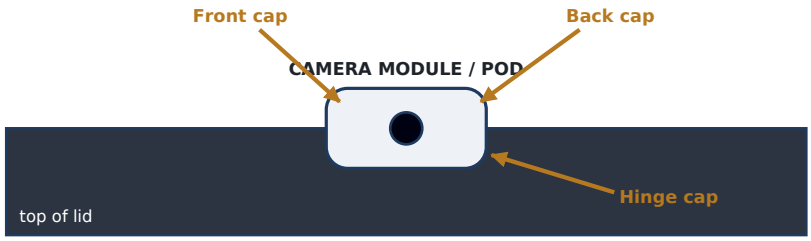
Does the camera connect through this cable? (Y/N) _____

Which hinge does the cable pass through? _____

R13 Camera module replacement

LEVEL 2

Fixes	Camera not detected, black image, broken lens
Part	Camera Module NB00430
Time	20 minutes
Tools	Plastic picks, tweezers



1. Bezel off first (it covers the camera's front).
2. Unplug the small camera cable with tweezers.
3. Lift the module by its board edges, never the lens.
4. Test in the Camera app before closing up.

Illustration: camera at the top-center of the lid. Cap shapes simplified.

Check first: S1, a powerwash (S3), and the Camera app in both pod positions. CTL's manual lists the camera as self-repairable.

Procedure

1. ☐ Do R3 and disconnect the battery.
2. ☐ Do R10 to remove the bezel.
3. ☐ Remove the camera caps needed to free the pod (R2). Record which ones.
4. ☐ Disconnect the camera cable with tweezers and lift the module out by its board edges.
5. ☐ Fit the new module in the same orientation and reconnect the cable.
6. ☐ Reconnect the battery and test the Camera app **in both pod positions** before refitting caps and bezel.

Verify before closing the ticket

- Clear image both ways.
- Pod rotates smoothly and holds position.
- Works in a Google Meet test call.

First-teardown notes — fill in the first time you do this repair

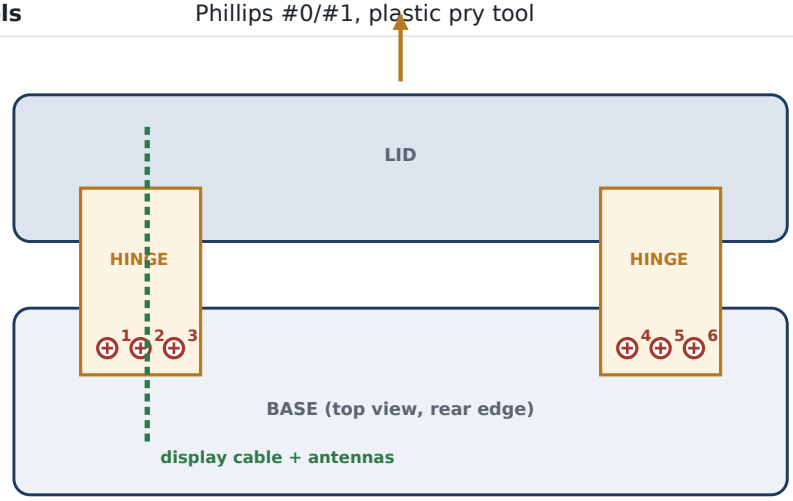
Caps removed to free the camera _____

Camera cable route _____

R14 Hinge replacement / removing the top half

LEVEL 2

Fixes	Floppy lid, cracked hinge, grinding. Removing the top half is also used for R8, R15, R17.
Part	Left Hinge or Right Hinge NL72/CT/L/NL73/CT
Time	30–45 minutes
Tools	Phillips #0/#1, plastic pry tool



1. Battery out, display cable unplugged at the mainboard.
2. Unthread cable and Wi-Fi antennas from the guides FIRST.
3. Remove the 6 hinge screws on the base side.
4. Support the lid, then lift it away from the base.

Illustration: hinge area seen from above with the D cover off. Positions approximate.

Check first: tighten the hinge screws. If a screw spins without tightening, the post is stripped. That's a cover replacement, not a hinge.

Procedure

1. ☐ Do R3 and remove the battery (CTL's prerequisite for removing the top half).
2. ☐ Disconnect the eDP/LVDS cable from the mainboard connector (R12 step 2).
3. ☐ Unthread the display cable and the Wi-Fi antennas from their cable guides. **CTL warns: do this before separating the halves.**
4. ☐ Remove the **six (6)** hinge screws.
5. ☐ Holding the lid so it can't fall, open it carefully and separate it from the bottom half.
6. ☐ To replace a hinge: remove the bezel (R10) and, if needed, the panel (R11) to reach the lid-side hinge screws. Swap the hinge. Left and right are mirror images.
7. ☐ Reassemble: seat the lid on the base, refit the 6 hinge screws, reroute cable and antennas through the guides, and reconnect.

Verify before closing the ticket

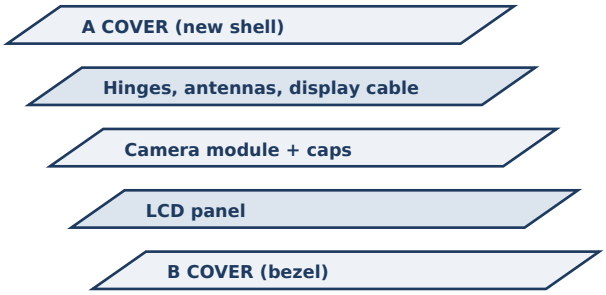
- Lid holds any angle and opens to 180°.
- Display steady through full travel.
- Wi-Fi strength normal (antennas not pinched).

First-teardown notes — fill in the first time you do this repair	
Lid-side hinge screw count	_____
Which hinge do the antennas pass through?	_____

R15 A cover (lid) replacement

LEVEL 3

Fixes	Cracked or broken lid back, stripped lid hinge posts
Part	NL72/CT and NL73/CT A cover
Time	45–60 minutes
Tools	Phillips #0/#1, plastic picks, tweezers, Kapton tape



1. Strip the old lid top to bottom: bezel, panel, camera, hinges, cable, antennas.
2. Build the new lid in the reverse order.
3. Test the display and Wi-Fi before the bezel goes on.

Illustration: lid layers, outside (top) to inside (bottom).

This moves every lid part to a new shell, combining R10–R14. Do each of those on its own first.

Procedure

1. ☐ Do R14 steps 1–5 to separate the lid from the base.
2. ☐ Do R10 (bezel), R11 steps 3–5 (panel), and R13 (camera and caps).
3. ☐ Remove the lid-side hinge screws and both hinges.
4. ☐ Lift out the display cable and Wi-Fi antennas, noting their route. Transfer any magnets, bumpers, and tape. Photograph first.
5. ☐ Build the new A cover in reverse: antennas, cable, hinges, camera, panel (test it), bezel.
6. ☐ Reattach the lid to the base (R14 step 7).

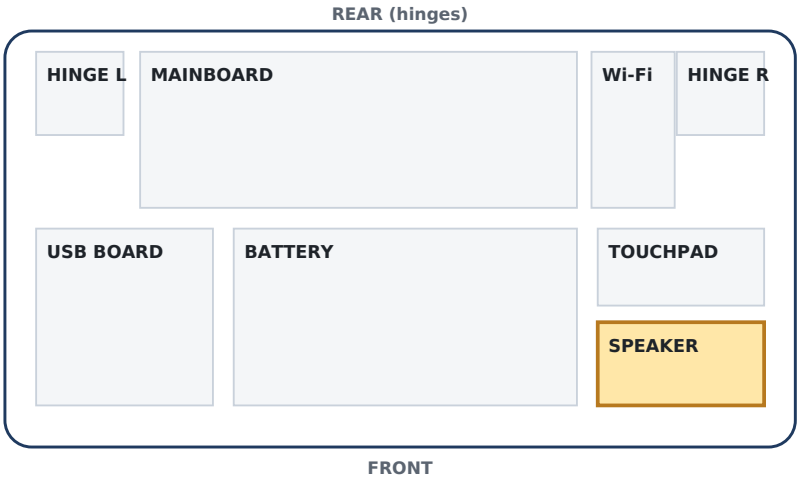
Verify before closing the ticket

- Display, camera, and Wi-Fi all work.
- Lid closes flush. Closing the lid puts the device to sleep (magnet in place).

R16 Speaker replacement

LEVEL 2

Fixes	No sound or crackling from one or both speakers after S1 and S3
Part	Renewed Left / Right Speaker for NL72 (confirm NL73 fit with CTL)
Time	20 minutes
Tools	Phillips #0, plastic tweezers



1. Battery disconnected.
2. Unplug the speaker cable at the mainboard.
3. Unroute the cable, noting each clip.
4. Lift the speaker; keep its rubber gasket.

Illustration: base interior, speaker highlighted. Position illustrative.

Check first: headphones work. If headphones also have no sound, the fault is software or mainboard, not the speakers.

Procedure

1. ☐ Do R3 and disconnect the battery.
2. ☐ Find the speakers and trace the speaker cable to the mainboard.
3. ☐ Disconnect the speaker cable at the mainboard.
4. ☐ Unroute the cable and lift the speaker(s) out. Note any rubber gaskets.
5. ☐ Fit the new speaker(s) with gaskets in place, route the cable the same way, and reconnect.
6. ☐ Reconnect the battery and test.

Verify before closing the ticket

- Clear sound from both sides with no rattle.
- Headphone jack still works.

First-teardown notes — fill in the first time you do this repair

Speaker screws or clips? _____

Are left and right separate parts? _____

R17 Top assembly (keyboard) replacement

LEVEL 3

Fixes	Multiple dead keys, a dead row, liquid-damaged keyboard, cracked palm rest
Part	NL73 Clamshell Series keyboard, NBKEYNL73 (NL73, NL73-8, NL73CT)
Time	60–90 minutes
Tools	Phillips #0/#1, plastic tools, tweezers, ESD strap, screw mat, Kapton tape

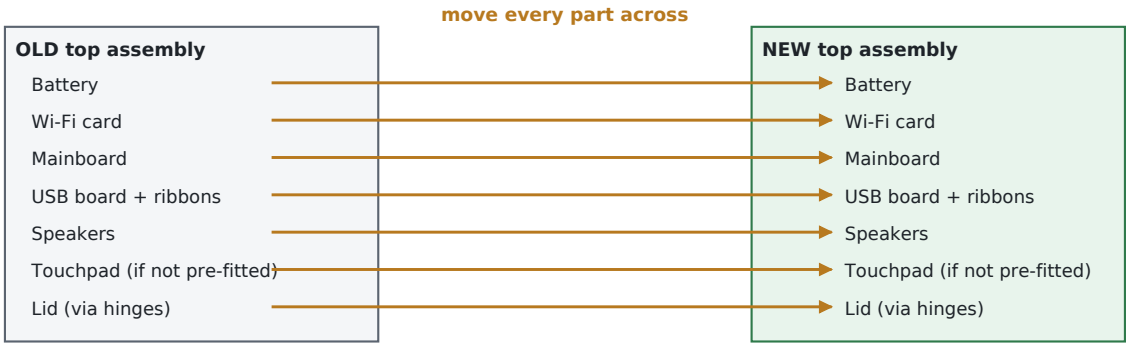


Illustration: what moves to the new top assembly. Check what comes pre-fitted first.

CTL's manual: the keyboard is part of the top assembly and can't be taken apart. If it fails, replace the whole top assembly. That means moving every part into the new one.

Check the box first. CTL's listing says "keyboard." Confirm when it arrives that it's the full top assembly (C cover with keyboard), and note what's pre-fitted (touchpad? speakers?).

Procedure

1. ☐ Do R3 and remove the battery (R4).
2. ☐ Remove the Wi-Fi card (R6). CTL lists this as a prerequisite for the mainboard.
3. ☐ Separate the lid from the base (R14 steps 2–5).
4. ☐ Remove the mainboard (R19 steps 3–7).
5. ☐ Remove the USB board (R8), speakers (R16), and touchpad (R7) if not pre-fitted on the new assembly.
6. ☐ Transfer everything to the new top assembly in reverse order: touchpad, speakers, USB board, mainboard, Wi-Fi, battery.
7. ☐ Reattach the lid (R14 step 7) and reconnect every cable, checking against your photos.
8. ☐ Reconnect the battery and test everything before refitting the D cover.

Verify before closing the ticket

- Every key passes the S2 keyboard test.
- Touchpad, display, camera, all ports, speakers, and Wi-Fi work.
- No leftover screws.

First-teardown notes — fill in the first time you do this repair

Parts pre-fitted on new assembly	_____
Total time for the first one	_____

R18 Liquid spill response

LEVEL 3

Fixes	Any liquid on or in the device
Part	Often none. Sometimes top assembly or mainboard.
Time	First response 10 min, drying 24–48 hours, cleaning 30–60 min
Tools	90%+ isopropyl alcohol, soft brush, lint-free cloths

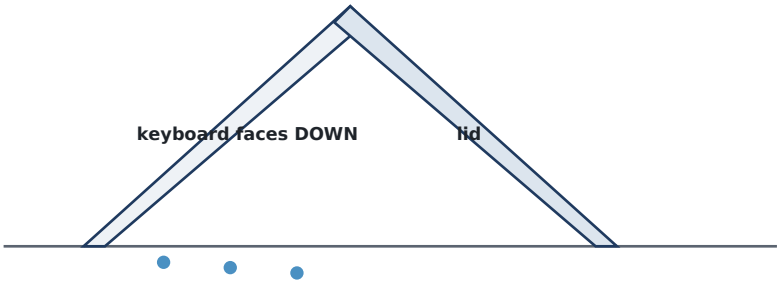


Illustration: drain position after a spill.

1. Power off (hold 10 s), unplug.
2. Tent upside down so liquid drains OUT.
3. Open and disconnect the battery ASAP.
4. Dry 24–48 h. No hair dryer, no rice.

The NL73 keyboard is spill-resistant, but that won't protect against a full cup. Speed matters.

Do not power on to test. Current through liquid causes the corrosion that kills boards.

Procedure

1. ☐ Hold Power for 10 seconds to force off. Unplug.
2. ☐ Tent the device upside down (lid at 90°, keyboard down) to drain it.
3. ☐ Do R3 and disconnect the battery as soon as possible.
4. ☐ Blot visible liquid. Don't wipe it around.
5. ☐ Sugary or sticky spills: clean affected areas with a brush and isopropyl alcohol.
6. ☐ Dry open, battery disconnected, for 24 hours minimum (48 if soaked). No hair dryer, no rice.
7. ☐ Inspect for white or green corrosion. Clean connectors (R5).
8. ☐ Reconnect the battery, plug in, and run S2 fully.

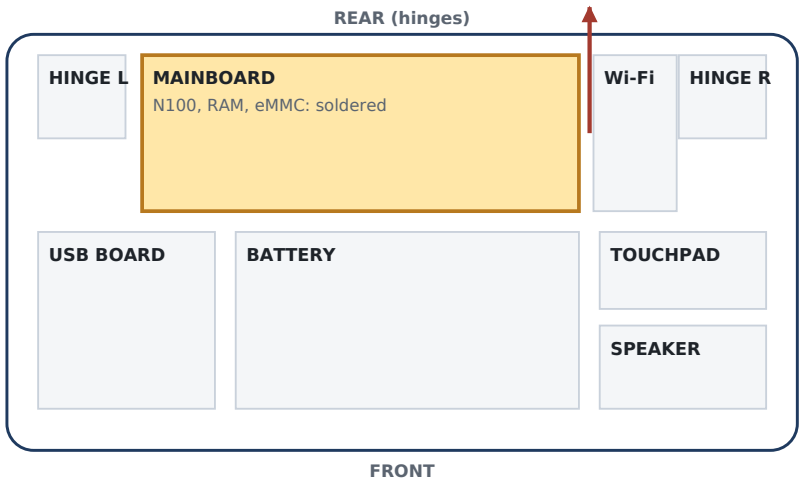
Verify before closing the ticket

- All S2 tests pass.
- Sticky or dead keys: R17.
- Corrosion on the mainboard: R19 or Sec. 6.

R19 Mainboard replacement

LEVEL 3

Fixes	Dead USB-C ports, no power with a good battery, failed eMMC, RAM errors, failed recovery
Part	Mainboard NL73 4/64, NB00489 • NL73-8 / NL73CT 8/64. **Match the RAM: NL73 = 4 GB, NL73-8 = 8 GB.**
Time	45–60 minutes
Tools	Phillips #0/#1, plastic tools, ESD strap



1. Battery and Wi-Fi card out first.
2. Unplug: 30/40-pin USB FFCs, touchpad FFC, keyboard FPC, speaker, LVDS (2-screw plate).
3. Lift from the RIGHT (Wi-Fi) side, per CTL.
4. Lift by the edges onto the ESD mat.

Illustration: base interior, mainboard highlighted. Red arrow = lift side.

Identity and enrollment. Storage and the device's identity live on the mainboard. Before swapping, the lead tech should update the asset record, deprovision the old device in the Google Admin console if needed, and plan to re-enroll. Ask CTL whether the replacement ships with the serial number set.

Confirm first with S1, S2, S3, and R5.

Procedure

1. ☐ Do R3 and remove the battery (R4).
2. ☐ Remove the Wi-Fi card (R6).
3. ☐ Disconnect the 30-pin and 40-pin USB board FFCs from the mainboard (unlock first).
4. ☐ Disconnect the touchpad FFC, keyboard FPC, and speaker cable.
5. ☐ Remove the **two (2)** screws on the LVDS protective plate, lift the locking arm, and remove the LVDS cable.
6. ☐ Check every cable is disconnected. **CTL's caution: disconnect all FFCs, FPCs, and cables first.**
7. ☐ Lift the mainboard from the **right (Wi-Fi) side** and remove it from the top assembly.
8. ☐ Fit the new board, then reverse steps 2–5 and refit the battery.
9. ☐ Power on. Run recovery (S3) if the board arrives blank, then enroll. Old board goes to e-waste.

Verify before closing the ticket

- Device boots, enrolls, and passes all of S2.
- Asset record and Admin console show the correct serial.

6 Out of scope: escalate

These need soldering or board-level work, or are explicitly not self-repairable per CTL's manual ("components not described in the disassembly section are not self-repairable"). Tag the device with the ticket and send it to the lead tech.

Fault	Why	Route
USB-C port on the mainboard loose or dead	Soldered to the mainboard	R19 swap or CTL RMA
Charging circuit failure (good battery, charger, USB board)	Mainboard components	R19 or CTL RMA
eMMC storage or RAM failure	Soldered	R19 or CTL RMA
Corrosion on the mainboard	Board-level cleaning or rework	R19 or CTL RMA
Burn marks, melted plastic, burnt smell	Fire hazard	Lead tech immediately
Stripped screw posts in top assembly or lid	Cover replacement or inserts	R15 / R17 or CTL
"Security module not working" during recovery	TPM/firmware fault	CTL support
Any repair needing a soldering iron or hot air	Program policy	CTL RMA

CTL contacts

- **CTL support:** 800-642-3087 • support@ctl.net • support.ctl.net
- **Parts:** ctl.net/collections/chromebook-parts
- **Manuals and videos:** support.ctl.net, "Chromebook Repair Videos and Manuals." Has an NL73 video folder and the NL72 manual (the NL73 reference). The NL73TW videos are for the convertible but show the same general technique.

7 Sources

Checked October 2026. Screw counts and step order are from CTL's NL72 manual. The procedures are the guide author's adaptation, not CTL-issued NL73 instructions.

- CTL, "NL72 Self-Repair Manual" (Google Doc linked from CTL support; CTL's designated reference for the NL73) — tools, pre-disassembly, screw counts, procedures, recovery messages.
- CTL Support, "Chromebook Repair Videos and Manuals" (updated Nov 20, 2025). support.ctl.net/hc/en-us/articles/11785795948307
- CTL, NL73 Series spec sheet (go.ctl.net/hubfs/Flyers/04.16_CTL_NL73_7-page_v03.pdf) — CPU, RAM, display, ports, wireless, weight, SKUs, update date.
- CTL, "Introducing the CTL Chromebook NL73 Series." ctl.net/pages/ctl-chromebook-nl73-series
- CTL, "CTL Genuine Chromebook Parts" catalog pages 1–11 — part names and compatibility. ctl.net/collections/chromebook-parts
- CTL product pages: NL73 battery (NBBNL73), NL73 Clamshell keyboard (NBKEYNL73), NL73 4/64 mainboard (NB00489), camera module (NB00430).